MIL-S-8932/3
12 August 1980
SUPERSEDING
MIL-S-8935
28 January 1965

MILITARY SPECIFICATION

SWITCHES, PRESSURE, AIRCRAFT, GAGE ,
TYPES I AND II, CLASS 3

This specification is l pproved for use by l ll Departments and Agencies of the Department of Defense.

The-complete requirements for procuring the switch assembly described herein shall consist of this document and the issue in effect of specification MIL-S-8932.

1. SCOPE

1*1 Scope- This specification covers the requirements for aircraft gage pressure switches for use in ac and dc applications.

1.2 Classification. Pressure switches covered by this specification shall be of the following types and class, as specified.

Type - Types I and II switches shall be as specified in MIL-S-8932.

Class - Class 3 switches shall be the aircraft gage pressure switches which respond to the difference in pressure between the ambient pressure and an applied pressure.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-8932 Switches, Pressure, Aircraft, General Specification for

Beneficial cements (recommendations, additions, deletions) an any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions . should be obtained from the procuring l ctivity or l s directed by the contracting officer).

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with the general specification MIL-S-8932 l nd l s specified herein. Each switch supplied under this specification shall incorporate l manufacturer's part number l nd shall have passed the first l rticle tests specified in MIL-S-8932.

3.1.1 Pressure switch port designations. Each gage pressure switch shall have two pressure ports as shown on the applicable MS. The designation "P" shall be stamped next to the higher pressure connection and the designation "V" shall be stamped next to the low pressure connection on the gage switch. Where one port is used for vent l nd case pressure release the port shall be marked "V + CPR". Where no case pressure release is required the port shall be marked "V".

3.2 Design.

3.2.1 Case. Case dimensions shall conform to the dimensions on the l pplicable MS.

3.2.2 Number of contacts. The number of contacts shall be as illustrated on the l pplicable figure 1 or 2.

3.2.3 Internal wiring. The switch contacts shall be wired to the connector terminals as illustrated on the applicable figure 1 and 2.

3.2.4 Electrical connector. The electrical connector used shall be l illustrated on the applicable figure 1 or 2.

3.2.5 Operating range. The range throughout which the switch must be l djustable l s specified on the applicable MS.

3.3 Current ratings l nd types of loads.

3.3.1 Alternating current (at)." The maximum value of l c current that the switch is required to carry shall be as specified on the l pplicable MS.

3.3.2 Direct current (dc). The maximum value of dc current that the switch is required to carry shall be as specified on the applicable MS.

3.3.3 Identification of product. A nameplate, permanently and legibly marked with the following information, shall be securely at-

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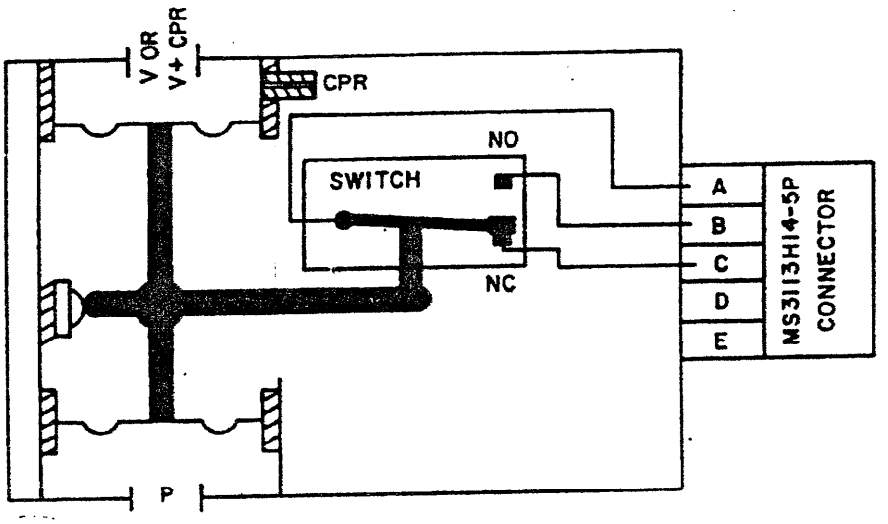


Figure 1. Gage pressure switch schematic, pressure port open.

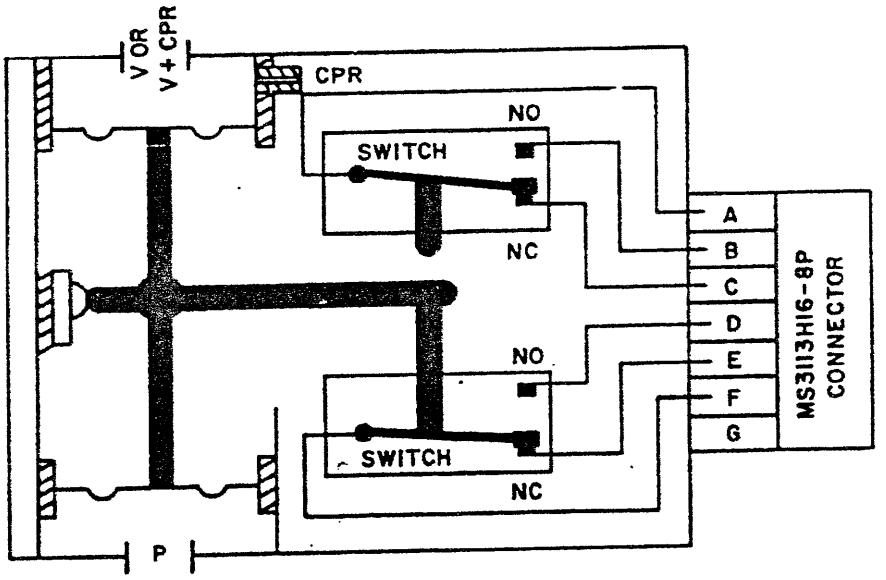


Figure 2. Gage pressure switch schematic, pressure port open.

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tached to the switch case. The nameplate shall not be attached to, or form l n integral part of a cover plate that is removable.

SWITCH, PRESSURE, AIRCRAFT
Type I {Engine mounted] or
Type II (Bulkhead mounted] (as applicable)
Class 3 - Gage
National Stock No.
MS Part No.
Manufacturer's Code and Part No.
Manufacturer's Serial No.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with the general specification MIL-S-8932 and as specified herein.

4.1.1 First article inspection. The first l rticle inspection shall consist of the applicable examinations and tests specified in MIL-S-8932 l nd as specified herein.

4.1.1.1 First article samples. The first article samples shall consist of three switches of the part number contracted for.

4.1.2 Quality conformance inspection. The quality conformance inspection shall consist of the tests specified in MIL-S-8932 and herein.

4.1.3 Tolerance. Tolerances for the tests and examinations specified in MIL-S-8932 and herein shall be as specified on the applicable MS.

4.1.4 Reference values. Before subjecting the pressure switch to any test requiring reference values, the reference values shall be obtained by the method specified in MIL-S-8932.

4.2 Proof pressure and drift. The test shall be conducted as specified in MIL-S-8932 except that the "V" or the "CPR" port shall have an ambient pressure equal to 14.5 ± 0.5 pounds per square inch (psi) while applying the normal system pressure and the proof pressure.

4.2.1 The test specified in 4.2 shall be repeated except that the "V" port shall be connected to a vacuum (less than 1 inch of mercury absolute) while applying the normal system pressure and the proof pressure.

4.3 Sensor balance. The following sequence of the test shall be conducted following the range of adjustment test. The test shall be performed in the order specified in 4.3.1.

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4.3.1 Sensor matching - gage switch. The gage switch pressure adjustment shall be set so that the switch actuates when the applied pressure is of a value equal to approximately 4/5 of the normal system pressure specified on the applicable KS. The "V" port shall be open to the atmosphere.

(a] A pressure shall be applied to the "P" port and the switch operated as specified in the procedure for operation of MIL-S-8932. The actuation and deactuation pressures shall be noted. The pressure difference shall be within the tolerance specified on the applicable MS.

(b] The test specified in (a) shall be repeated except that a pressure equal to approximately 15 psia shall be applied to the "V" port. The difference between the actuating and deactuating pressures shall be noted. The difference between them and the vent pressure shall not differ from the values noted in (a) by more than the tolerance specified on the l pplicable MS.

(c) The test specified in (b) shall be repeated except that 14 inches of mercury absolute shall be supplied to the "V" port.

(d) The test specified in (b) shall be repeated except that 10 inches of mercury absolute shall be applied to the "V + CPR" port.

(e) The test specified in (b) shall be repeated except that 5 inches of mercury absolute shall be applied to t-he "V + CPR" port.

5. PACKAGING

5.1 " Preparation for delivery. preparation for delivery shall be in accordance with MIL-S-8932.

6. NOTES

6.1 Intended use. The aircraft gage pressure switches are intended for use in aircraft.

6.2 Ordering data. The instructions and definitions contained in Section 6 of MIL-S-8932 are applicable to, and form a part of this specification.

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